PULLBACK ALONG A FLAT MORPHISM OF RELATIVE DIMENSION R WHOSE FIBRES ARE AFFINE R-SPACES IS SURJECTIVE ON CHOW GROUPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a decomposition into locally finite images reduced to an integral base, an induction on the dimension of the base through an open subspace and the complement, and a final descent of the generic fibre isomorphism

Editorial note. The label lemma-pullback-affine-fibres-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chow groups and rational equivalence for locally finite sums of cycles, together with the scheme theoretic image of an integral closed subspace under the morphism
- The exact sequences relating the Chow groups of a space, an open subspace and the preimage of its complement, and the lifting of cycles from an open subspace to the whole space
- The descent of an isomorphism of finite presentation from the generic point of an integral scheme to a nonempty open subspace, and the scheme case of the statement

Source reference. Original lemma 02TT: sources/stacks-chow/tags.txt, source lines 3708–3708. Prerequisite pointer invoked at proof line(s) 4013; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RC: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1306–1306. Prerequisite pointer invoked at proof line(s) 3995; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EEG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof line(s) 3963; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02RX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1951–1951. Prerequisite pointer invoked at proof line(s) 4000; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RZ: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1991–1991. Prerequisite pointer invoked at proof line(s) 3983; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 3949; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07SK: sources/stacks-chow/tags.txt, source lines 9905–9905. Prerequisite pointer invoked at proof line(s) 4011; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good. Let $f : X \to Y$ be a quasi-compact flat morphism over B of relative dimension r. Assume that for every $y \in Y$ we have $X_y \cong \mathbf{A}^r_{\kappa(y)}$. Then $f^* : \mathrm{CH}_k(Y) \to \mathrm{CH}_{k+r}(X)$ is surjective for all $k \in \mathbf{Z}$.*

Proof. Let $\alpha \in \mathrm{CH}_{k+r}(X)$. Write $\alpha = \sum m_j[W_j]$ with $m_j \neq 0$ and W_j pairwise distinct integral closed subspaces of δ-dimension $k + r$. Then the family $\{W_j\}$ is locally finite in X. Let $Z_j \subset Y$ be the integral closed subspace such that we obtain a dominant morphism $W_j \to Z_j$ as in Lemma 0EEG. For any quasi-compact open $V \subset Y$ we see that $f^{-1}(V) \cap W_j$ is nonempty only for finitely many j. Hence the collection Z_j of closures of images is a locally finite collection of integral closed subspaces of Y.

Consider the fibre product diagrams

$$\begin{array}{ccc} f^{-1}(Z_j) & \longrightarrow & X \\ {\scriptstyle f_j}\downarrow & & \downarrow{\scriptstyle f} \\ Z_j & \longrightarrow & Y \end{array}$$

Suppose that $[W_j] \in Z_{k+r}(f^{-1}(Z_j))$ is rationally equivalent to $f_j^*\beta_j$ for some k-cycle $\beta_j \in \mathrm{CH}_k(Z_j)$. Then $\beta = \sum m_j\beta_j$ will be a k-cycle on Y and $f^*\beta = \sum m_j f_j^*\beta_j$ will be rationally equivalent to α (see Remark 02RZ). This reduces us to the case Y integral, and $\alpha = [W]$ for some integral closed subscheme of X dominating Y. In particular we may assume that $d = \dim_\delta(Y) < \infty$.

Hence we can use induction on $d = \dim_\delta(Y)$. If $d < k$, then $\mathrm{CH}_{k+r}(X) = 0$ and the lemma holds; this is the base case of the induction. Consider a nonempty open $V \subset Y$. Suppose that we can show that $\alpha|_{f^{-1}(V)} = f^*\beta$ for some $\beta \in Z_k(V)$. By Lemma 02RC we see that $\beta = \beta'|_V$ for some $\beta' \in Z_k(Y)$. By the exact sequence $\mathrm{CH}_k(f^{-1}(Y \setminus V)) \to \mathrm{CH}_k(X) \to \mathrm{CH}_k(f^{-1}(V))$ of Lemma 02RX we see that $\alpha - f^*\beta'$ comes from a cycle $\alpha' \in \mathrm{CH}_{k+r}(f^{-1}(Y \setminus V))$. Since $\dim_\delta(Y \setminus V) < d$ we win by induction on d.

In particular, by replacing Y by a suitable open we may assume Y is a scheme with generic point η. The isomorphism $Y_\eta \cong \mathbf{A}^r_\eta$ extends to an isomorphism over a nonempty open $V \subset Y$, see Limits of Spaces, Lemma 07SK. This reduces us to the case of schemes which is Chow Homology, Lemma 02TT. $\square$